Homework 0: Qubits and Quantum Gates

Quantum Computing Training 2026

This homework assignment is written so that you can work on it before seeing the lecture slides. It introduces the notation and facts you need along the way. The goal is to build comfort with three core ideas:

- A qubit is described by complex amplitudes attached to the basis states $|0\rangle$ and $|1\rangle$;
- Measurement probabilities come from squared magnitudes of amplitudes;
- Quantum gates act on qubits by matrix multiplication.

The first lecture will go further into historical context, qubits, gates, entanglement, and teleportation. This problem set focuses on qubits, the Bloch sphere, gates, and a short Bell-state bonus. Teleportation and historical context are intentionally omitted.

Useful notation. The symbol i means $\sqrt{-1}$. The notation $|0\rangle$ and $|1\rangle$ is called ket notation. For this problem set, you can think of

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A general single-qubit state is a linear combination of these two basis states.

Problem 1: Getting Comfortable with Single Qubits

A classical bit is the basic unit of information in ordinary digital computing. At any moment, a classical bit has one of two possible values: 0 or 1. A qubit is the basic unit of information in quantum computing. When we measure a qubit in the computational basis, the measurement result is also an ordinary classical bit: either 0 or 1. Before measurement, the qubit state is described using two complex numbers called amplitudes, one attached to the basis state $|0\rangle$ and one attached to the basis state $|1\rangle$. These amplitudes determine the probabilities of the two possible measurement outcomes.

A single qubit can be written as

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,,$$

where $\alpha, \beta \in \mathbb{C}$ and the normalization condition is

$$|\alpha|^2 + |\beta|^2 = 1.$$

Here α is the amplitude of $|0\rangle$ and β is the amplitude of $|1\rangle$. The squared magnitudes $|\alpha|^2$ and $|\beta|^2$ give the probabilities of measuring outcomes 0 and 1, respectively, when the qubit is measured in the computational basis $\{|0\rangle, |1\rangle\}$. The normalization condition says that the total probability across all possible measurement outcomes is 1.

For a complex number $a + b\mathrm{i}$, its squared magnitude is

$$|a + b\mathrm{i}|^2 = a^2 + b^2.$$

For example, $|4\mathrm{i}/5|^2 = 16/25$ and $|-1/\sqrt{2}|^2 = 1/2$.

(a) For each proposed state below, determine whether it is a valid qubit state. If it is not valid, explain why.

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \qquad |\psi_2\rangle = \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle, \qquad |\psi_3\rangle = \frac{3}{5}|0\rangle + \frac{4\mathrm{i}}{5}|1\rangle.$$

(b) Suppose

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

If we measure this qubit in the computational basis $\{|0\rangle, |1\rangle\}$, what are the probabilities of getting outcome 0 and outcome 1?

(c) Imagine that we prepare 1000 identical copies of the state from part (b) and measure all of them in the computational basis. About how many 0 outcomes and how many 1 outcomes should we expect? Why should we not expect exactly those numbers every time?

Then check this prediction using Qiskit. Create a one-qubit circuit that prepares the state

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle,$$

measure the qubit in the computational basis, run the circuit for 1000 shots, and plot a histogram of the output counts. Your histogram should have bars for the measured bit values 0 and 1. Compare the simulated counts with the probabilities from part (b).

(d) Consider the two states

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

If both are measured in the computational basis, do they give different measurement probabilities? Explain what information is lost if we only look at computational-basis measurement probabilities.

Problem 2: The Bloch Sphere Representation

The Bloch sphere is a geometric way to draw any pure state of one qubit as a point on the surface of a sphere. The north pole represents $|0\rangle$ and the south pole represents $|1\rangle$. Points around the equator represent equal-probability superpositions of $|0\rangle$ and $|1\rangle$ with different relative phases.

Any single-qubit pure state can be written, up to an overall physically irrelevant global phase, as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{\mathrm{i}\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle,$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

The angle θ measures how far the state is from the north pole $|0\rangle$ toward the south pole $|1\rangle$. The angle ϕ measures direction around the equator and controls the relative phase between the $|0\rangle$ and $|1\rangle$ parts of the state. The measurement probabilities in the computational basis are

$$P(0) = \cos^2\left(\frac{\theta}{2}\right), \qquad P(1) = \sin^2\left(\frac{\theta}{2}\right).$$

(a) Identify the values of θ and ϕ for the computational basis states $|0\rangle$ and $|1\rangle$. Is ϕ uniquely determined for both of these states?

(b) Find θ and ϕ for each of the following states:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \qquad |+i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + \mathrm{i}\,|1\rangle).$$

(c) For the state

$$|\psi\rangle = \cos\left(\frac{\pi}{6}\right)|0\rangle + e^{\mathrm{i}\pi/3}\sin\left(\frac{\pi}{6}\right)|1\rangle,$$

compute the probabilities of measuring 0 and 1 in the computational basis.

(d) The Bloch sphere coordinates corresponding to (θ, ϕ) are

$$x = \sin\theta\cos\phi, \qquad y = \sin\theta\sin\phi, \qquad z = \cos\theta.$$

Compute (x, y, z) for the states $|0\rangle$, $|1\rangle$, $|+\rangle$, and $|+i\rangle$.

(e) In your own words, explain the difference between changing θ and changing ϕ. Which one changes the measurement probabilities in the computational basis, and which one changes only the relative phase between $|0\rangle$ and $|1\rangle$?

Problem 3: Quantum Gates as Matrix Operations

Quantum gates are operations that change qubit states. In this problem, each gate is represented by a matrix, and each qubit state is represented by a column vector. Applying a gate means multiplying the gate matrix by the state vector.

In the computational basis, we represent

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Some important single-qubit gates are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The gate X is often called the bit-flip gate, Z is often called the phase-flip gate, and H is called the Hadamard gate.

(a) Compute $X|0\rangle$, $X|1\rangle$, $Z|0\rangle$, and $Z|1\rangle$. Based on your calculation, describe in words what the X gate and Z gate do to the computational basis states.

(b) Compute $H|0\rangle$ and $H|1\rangle$. Give your answers using $|0\rangle$ and $|1\rangle$ notation.

(c) Starting from $|0\rangle$, apply the following sequence of gates:

$$|0\rangle \xrightarrow{H} \xrightarrow{Z} \xrightarrow{H} .$$

What final state do you get? This sequence is often summarized as the identity

$$HZH = X.$$

Does your result agree with that identity for the input state $|0\rangle$?

(d) Now start from a general qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle .$$

Compute $X|\psi\rangle$ and $Z|\psi\rangle$. Which gate swaps amplitudes, and which gate changes a relative sign?

(e) A common beginner mistake is to say that the Z gate "does nothing" because measuring $Z|0\rangle$ or $Z|1\rangle$ gives the same computational-basis outcomes as before. Use the state $|+\rangle$ to show why this is not the full story.

Bonus: A First Look at Bell States

Two qubits together have four computational-basis states:

$$|00\rangle, \qquad |01\rangle, \qquad |10\rangle, \qquad |11\rangle .$$

The first slot labels the first qubit, and the second slot labels the second qubit. For example, $|01\rangle$ means that the first qubit is in state 0 and the second qubit is in state 1.

The Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

is a simple example of an entangled two-qubit state. Its amplitudes are attached only to $|00\rangle$ and $|11\rangle$, so those are the only two computational-basis outcomes with nonzero probability.

(a) If we measure both qubits of $|\Phi^+\rangle$ in the computational basis, what outcomes can occur, and with what probabilities?

(b) Suppose the first qubit is measured and the outcome is 0. What must the second qubit be if we immediately measure it in the computational basis?

(c) Suppose the first qubit is measured and the outcome is 1. What must the second qubit be if we immediately measure it in the computational basis?

(d) Explain why this state has stronger correlations than two independent qubits that each have a 50% chance of being measured as 0 or 1.

(e) Check your answer using Qiskit. Build a two-qubit circuit that prepares $|\Phi^+\rangle$ from the initial state $|00\rangle$ by applying a Hadamard gate to the first qubit and then a controlled-X gate, also called a CNOT gate, from the first qubit to the second qubit. Measure both qubits, run the circuit for 1000 shots, and plot a histogram of the two-bit output strings. Which output strings appear most often? How does the histogram show the correlation between the two qubits?